Laws of motion CHEAT SHEET

THE THREE LAWS

1 no net force, no change in velocity

cruising car: engine push = drag → v stays

puck at 4 m/s: μk 0.05 stops in 16.33 m; μk 0: never

2 $F_{net} = \frac{\Delta p}{\Delta t} = m\,a$ (a along F)

3 forces come in pairs, on two different bodies

book on a table: W and N are NOT a pair (both on the book)

inertia: mass resists change · mass in kg

CONSERVATION OF MOMENTUM

conservation (no outside net force):

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

recoil: skaters 60 kg and 40 kg → −2 m/s and +3 m/s

probe 1,000 kg throws 2 kg at 3,000 m/s → +6.01 m/s

coupling: 30,000 kg at 2 m/s + 20,000 kg → 1.2 m/s

MOMENTUM, FORCE, WEIGHT, IMPULSE

momentum $p = m\,v$: 0.45 kg at 20 m/s → 9 kg m/s

1 N = 1 kg × 1 m/s²: 1,200 kg at 2 m/s² → 2,400 N

weight $W = m\,g$: 60 kg → 588 N (Moon: 97.2 N)

impulse $F\,\Delta t = \Delta p$ (area under F–t, N s)

catch 0.16 kg at 25 m/s: Δp = 4 N s;

0.01 s → 400 N · 0.02 s → 200 N (simulated)

kick back: Δp = 0.45 × (18 − (−12)) = 13.5 N s → 1,350 N

edges: F = 0 → a = 0; Δt → 0 → F without limit

FREE-BODY DIAGRAMS: FOUR STEPS

1 pick ONE body; draw it cut free

2 every force ON it: W down, N at right angles

to the surface, tension along a rope, friction along it

3 split each into x and y parts

4 $\Sigma F_x = m\,a_x$, $\Sigma F_y = m\,a_y$

crate 5 kg, rope 20 N at 30°: 17.32 N along, 10 N up;

$N = 49 - 10 = 39$ N; $a = 3.46$ m/s²

EQUILIBRIUM, LIFTS, CRATES

equilibrium: $\Sigma F_x = 0$ and $\Sigma F_y = 0$

sign 10 kg, two cables at 30°: T = 98 N each

30° and 60°: 49 N and 84.87 N; at 5°: 562 N

lift: $N - m\,g = m\,a$ (scale reads N)

60 kg: +1.2 → 660 N · −1.2 → 516 N · free fall → 0

crates 4 + 2 kg pulled by 30 N: a = 5 m/s², link 10 N

Laws of motion CHEAT SHEET

FRICTION

static: $f \le \mu_s N$ (only as much as it must)

kinetic: $f = \mu_k N$, usually $\mu_k < \mu_s$

box 40 kg: N 392 N; μ_s 0.5 → starts at 196 N; μ_k 0.4 → 156.8 N

laws: proportional to N; not the area of contact

rolling friction: far smaller · lubrication: keep apart

ON A SLOPE

slips when $\tan\theta > \mu_s$: μ_s 0.5 → 26.57°

sliding: $a = g(\sin\theta - \mu_k \cos\theta)$

35°, μ_k 0.4 → 2.41 m/s²; 4.82 m in 2 s (simulated)

backwards: $\mu_k = (g \sin\theta - a)/(g \cos\theta)$: 30°, 2.4 → 0.29

edges: $\mu = 0 \to g \sin\theta$; $\theta = 90° \to g$

MOVING IN A CIRCLE

$a = \frac{v^2}{r}$ to the centre; $F = \frac{m v^2}{r}$ (a job, not a new force)

ball 0.2 kg, 0.8 m, 4 m/s: 4 N; period 1.26 s

level road: $v \le \sqrt{(\mu_s r g)}$ (mass cancels)

r 30 m: dry 0.7 → 51.6 km/h · ice 0.1 → 19.5 km/h

banking, no friction: $\tan\theta = \frac{v^2}{r g}$

track r 25 m at 15 m/s → 42.56°

BANKED ROAD WITH FRICTION

$v^2_{max} = r g \frac{\tan\theta + \mu s}{1 - \mu s \tan\theta}$ (friction downhill)

$v^2_{min} = r g \frac{\tan\theta - \mu s}{1 + \mu s \tan\theta}$ (0 if $\tan\theta \le \mu s$)

r 40 m, 10°, dry 0.6: 0 to 66.4 km/h

icy 0.1: 19.5 to 37.8 km/h

edges: $\mu s = 0$ → one speed (29.9 km/h); $\theta = 0$ → level road

exit ramp r 60 m, 50 km/h: 18.16° needs no grip,

but on ice (μs 0.1) cars under 41 km/h slide in

$\tan\theta = 0.08$ → 4.57°: a stopped car holds;

top speeds dry 79.4 · wet 61.5 · icy 37.2 km/h

CHECK BEFORE YOU TRUST IT

before you trust an answer:

1 the body alone, with every force acting on it?

2 added as arrows, a sign for each direction?

3 the edges: no friction, zero angle, zero speed?

Python: angles in radians! tan(radians(30)) = 0.577,

tan(30) = −6.41; degrees(atan(0.5)) = 26.57